

Mathematical Foundations of Reinforcement Learning and Online Decision Making Homework 1

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1 Solution to Question 1

1.1 (a)

Definition 1.1.1. A Markov chain with state space $\mathcal{X}$ is irreducible if for any $x, y \in \mathcal{X}$, there exists $t \geq 1$ such that $P^t(x, y) > 0$.

Definition 1.1.2. The period of a state $x \in \mathcal{X}$ is defined as $d(x) := \gcd \mathcal{T}(x)$, where

$$\mathcal{T}(x) := \{t \geq 1 : P^t(x, x) > 0\}.$$

Definition 1.1.3. A Markov chain is aperiodic if it is irreducible and $d(x) = 1$ for all $x \in \mathcal{X}$.

Lemma 1.1.1. In an irreducible Markov chain, $d(x) = d(y)$ for all $x, y \in \mathcal{X}$.

Proof. Let $x, y \in \mathcal{X}$. By Definition 1.1.1, there exist non-negative integers $r, l \geq 1$ such that $P^r(x, y) > 0$ and $P^l(y, x) > 0$.

Let $m = r + l$. By using properties of matrix multiplication and assuming that $\mathcal{X}$ is finite, we can write:

For path $x \rightarrow y \rightarrow x$:

$$P^m(x, x) \geq P^r(x, y)P^l(y, x) > 0 \implies m \in \mathcal{T}(x).$$

For path $y \rightarrow x \rightarrow y$:

$$P^m(y, y) \geq P^l(y, x)P^r(x, y) > 0 \implies m \in \mathcal{T}(y).$$

Now, for any $t \in \mathcal{T}(x)$, we have $P^t(x, x) > 0$, since we have a *self-loop*. Considering the path $y \rightarrow x \rightarrow x \rightarrow y$, we obtain:

$$P^{t+m}(y, y) \geq P^l(y, x)P^t(x, x)P^r(x, y) > 0 \implies t + m \in \mathcal{T}(y).$$

Since $d(y) = \gcd \mathcal{T}(y)$, $d(y)$ divides every element of $\mathcal{T}(y)$. In particular, $d(y) \mid m$ and $d(y) \mid (t + m)$. Therefore, $d(y)$ must divide $(t + m) - m = t$.

Because $d(y)$ divides every $t \in \mathcal{T}(x)$, it must also divide $\gcd \mathcal{T}(x) = d(x)$. Thus, $d(y) \leq d(x)$.

By symmetry, reversing the roles of x and y yields $d(x) \leq d(y)$. Hence, $d(x) = d(y)$ for all $x, y \in \mathcal{X}$. $\square$

Main Result: Suppose the irreducible Markov chain contains a state $x \in \mathcal{X}$ with a self-loop, i.e., $P(x, x) > 0$.

Then $1 \in \mathcal{T}(x)$ by construction, which implies $d(x) = \gcd \mathcal{T}(x) = 1$ (using number theory). By the lemma 1.1.1, every state $y \in \mathcal{X}$ has period $d(y) = d(x) = 1$.

Therefore, by definition 1.1.3, the given Markov chain is *aperiodic*.

1.2 (b)

Given the markov chain is *irreducible* and *aperiodic*.

So, $x, y \in \mathcal{X}$, there exists $t \geq 1$ such that $P^t(x, y) > 0$ and $d(x) = 1$ for all $x \in \mathcal{X}$. Since there are no self-loops, all the diagonal elements of P is 0.

We consider a markov chain with state $x \in \mathcal{X}$, where $|\mathcal{X}| = 3$. P is stochastic, with row sums = 1.

Let P be defined as:

$$\begin{bmatrix} 0 & a & 1-a \\ b & 0 & 1-b \\ c & 1-c & 0 \end{bmatrix}$$

where $a, b, c \in (0, 1]$.

By construction, P satisfies irreducibility since $P^t(x, y) > 0$ for $x \in \mathcal{X}$, and $t > 1$ (no self-loops).

With the above conditions, we can take $a = \frac{1}{2}$, $b = \frac{1}{3}$ and $c = \frac{1}{4}$.

Thus, after substituting a, b, c in P for a irreducible and aperiodic markov chain is below:

$$\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{1}{4} & \frac{3}{4} & 0 \end{bmatrix}$$

2 Solution to Question 2

Let P be an irreducible transition matrix of a markov chain with period $d(x) = b$.

2.1 (a)

Lets say $x, y, z \in \mathcal{X}$. Since $d(x) = b$, for a $m \geq 1$ and $P^m(z, z) > 0$, from Definition 1.1.1 and 1.1.2:

$$m \equiv 0 \pmod{b}$$

In the question, the given condition is:

$$i + 1 \equiv 0 \pmod{b}$$

If $z \in C_1$, then z needs m to satisfy $m \equiv 0 \pmod{b}$ to return to itself, since b is the period, by Definition 1.1.2.

Now, with our assumption that $C_i \rightarrow C_{i+1}$ takes 1 transition step, going step-by-step through induction, based on the number of steps taken from z to:

Set C_1 : Contains z and any state reachable from z in m steps where $m \equiv 0 \pmod{b}$.

Set C_2 : Taking 1 step from any state in C_1 leads to a state reachable from z in $m + 1$ steps, where $(m + 1) \equiv 1 \pmod{b}$.

Set C_3 : Taking another step leads to states reachable from z in $m + 2$ steps, where $(m + 2) \equiv 2 \pmod{b}$.

By continuing this process for $i - 1$ steps, we define the partition set C_i for $i \in \{1, 2, \dots, b\}$: a state $x \in C_i$ if there exists a path from $z \rightarrow x$ of length $m + i - 1$ steps with:

$$m \equiv i - 1 \pmod{b}$$

Proof. Now consider a single transition $P(x, y) > 0$ with $x \in C_i$.

Since $x \in C_i$, the path length from $z \rightarrow x$ satisfies $m \equiv i - 1 \pmod{b}$.

Moving from $x \rightarrow y$ takes 1 additional step, given $P(x, y) > 0$ in the question, making the path length from $z \rightarrow y$ equal to $m + 1$.

Evaluating this new length modulo b :

$$(m + 1) \equiv (i - 1) + 1 \equiv i \pmod{b}$$

Since the remainder is i , the state y belongs to class C_{i+1} .

With index addition taken modulo b , so $C_b \rightarrow C_1$ (circular behavior). $\square$

This completes the proof that $P(x, y) > 0$ only when transitions move from C_i to C_{i+1} for given condition $i + 1 \equiv 0 \pmod{b}$, which means for $P(x, y) > 0$, $x \in C_i$ and $y \in C_{i+1}$.

2.2 (b)

Given condition:

$$|\mathcal{X}| = n$$

$$n_i = |\mathcal{C}_i|$$

$$N_i = \sum_{j=1}^i n_j$$

$$\text{for } i = 2, \quad n_1 + n_2 + \dots + n_{i-1} = \sum_{j=1}^{i-1} n_j = n_1$$

So the general term of set C_i is:

$$C_i = \{x_{N_{i-1}+1}, \dots, x_{N_i}\}$$

Lemma 2.2.1. *Since, $P(x, y) > 0$, $x \in C_i$ and $y \in C_{i+1}$ is proved above, $P(x, y) = 0$ for $x \notin C_i$ or $y \notin C_{i+1}$. This is by Proof 2.1.*

If C_i is considered a set of $x \in \mathcal{X}$, then we can consider $\{C_1, \dots, C_b\}$ as the set of states for the same P .

Originally, $P \in \mathbb{R}^{n \times n}$. But while considering $\{C_1, \dots, C_b\}$ as the set of states for the same P , we can re-write $P \in \mathbb{R}^{b \times b}$, where:

$$P = \begin{bmatrix} \mathbf{0} & P_{1,2} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & P_{2,3} & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & P_{b-1,b} \\ P_{b,1} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix}$$

Most of the cells in above P is 0 due to Lemma 2.2.1.

Now, each $P_{i,i+1} \in \mathbb{R}^{n_i, n_{i+1}}$.

Total elements in P :

$$|P| = n^2$$

Total elements in $P_{i,i+1}$:

$$|P_{i,i+1}| = P_{(i \bmod b), (i+1 \bmod b)} = n_i \times n_{i+1}$$

Total non-zero elements in P :

$$\begin{aligned} & \sum_{i=1}^b P_{(i \bmod b), (i+1 \bmod b)} \\ \implies & \sum_{i=1}^{b-1} n_i \times n_{i+1} + (n_b \times n_1) \end{aligned}$$

Total zero elements in P :

$$\begin{aligned} & \frac{n^2 - \sum_{i=1}^{b-1} n_i \times n_{i+1} - (n_b \times n_1)}{n^2} \\ \implies & 1 - \left(\frac{\sum_{i=1}^{b-1} n_i \times n_{i+1} + (n_b \times n_1)}{n^2} \right) \\ \implies & 1 - \left(\frac{(n_1 \cdot n_2) + (n_2 \cdot n_3) + \dots + (n_b \cdot n_1)}{n^2} \right) \end{aligned}$$

The above forms looks very similar to dot product between two vectors, whose elements are shifted by one index.

If $\mathbf{a} = (n_1, n_2, \dots, n_b)$ and $\mathbf{b} = (n_2, n_3, \dots, n_b, n_1)$, then:

$$\mathbf{a} \cdot \mathbf{b} = (n_1 \cdot n_2) + (n_2 \cdot n_3) + \dots + (n_b \cdot n_1)$$

Cauchy-Schwarz inequality for two vectors $\mathbf{a}$ and $\mathbf{b}$ says that:

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \cdot \|\mathbf{b}\| \quad \text{for } a, b \in \mathbb{R}^n$$

Based on this inequality, we can say that:

$$\begin{aligned} (n_1 \cdot n_2) + (n_2 \cdot n_3) + \dots + (n_b \cdot n_1) &\leq \sqrt{\left(\sum_{i=1}^b n_i^2\right)} \sqrt{\left(\sum_{i=1}^b n_i^2\right)} = \left(\sum_{i=1}^b n_i^2\right) \\ \implies (n_1 \cdot n_2) + (n_2 \cdot n_3) + \dots + (n_b \cdot n_1) &\leq \left(\sum_{i=1}^b n_i^2\right) \\ \implies \frac{(n_1 \cdot n_2) + (n_2 \cdot n_3) + \dots + (n_b \cdot n_1)}{n^2} &\leq \frac{\left(\sum_{i=1}^b n_i^2\right)}{n^2} \end{aligned}$$

From the definition and given $|\mathcal{X}| = n$, $\sum_{i=1}^b n_i = n$.

To characterize the minimum of the $\sum_{i=1}^b n_i = n$, we can use Lagrange multiplier, since this is a multi-variable constraint based optimization problem.

Let $f(n_1, n_2, \dots, n_b) = \sum_{i=1}^b n_i^2$

And constraint $g(n_1, n_2, \dots, n_b) = \sum_{i=1}^b n_i - n = 0$.

So, using Lagrange Multiplier:

$$L(n_1, n_2, \dots, n_b, \lambda) = \sum_{i=1}^b n_i^2 - \lambda \left(\sum_{i=1}^b n_i - n \right)$$

To find the critical point of L , we can take partial derivative for any n_i :

$$\frac{\partial L}{\partial n_i} = 2n_i - \lambda = 0 \implies n_i = \frac{\lambda}{2}$$

This means, for the f to reach minimum, all n_i should be equal. Since, $n_1 = n_2 = \dots = n_b$ and $\sum_{i=1}^b n_i = n$, for minimum f , $n_i = \frac{n}{b}$, since $n, b > 0$.

Now, since $\sum_{i=1}^b n_i = n$, and $n_1, n_2, \dots, n_b > 0$, it implies $n_1, n_2, \dots, n_b \leq n$.

By this inequality and conditions, if one of the $n_i = n - b + 1$ and other $n_i = 1$, then we still have $\sum_{i=1}^b n_i = n$, while $\sum_{i=1}^b n_i^2 = (n - b + 1)^2 + (b - 1) < n^2$ being the maximum for f .

But since

Hence, we can see from above minimum and maximum that:

$$\frac{n^2}{b} \leq \sum_{i=1}^b n_i^2 \leq (n - b + 1)^2 + (b - 1) < n^2$$

With re-arranging inequality and Cauchy-Schwarz inequality:

$$\frac{n^2}{n^2b} \leq \frac{(n_1.n_2) + (n_2.n_3) + \dots + (n_b.n_1)}{n^2} \leq \frac{\sum_{i=1}^b n_i^2}{n^2} \leq \frac{(n-b+1)^2 + (b-1)}{n^2} < \frac{n^2}{n^2}$$

$$\frac{1}{b} \leq \frac{(n_1.n_2) + (n_2.n_3) + \dots + (n_b.n_1)}{n^2} \leq \frac{(n-b+1)^2 + (b-1)}{n^2} < 1$$

$$\frac{1}{b} \leq \frac{(n_1.n_2) + (n_2.n_3) + \dots + (n_b.n_1)}{n^2} \leq \frac{(n-b+1)^2 + (b-1)}{n^2}$$

If we subtract it from 1 for getting zero elements in P and reverse the sign of inequality:

$$1 - \frac{(n-b+1)^2 + (b-1)}{n^2} < 1 - \frac{(n_1.n_2) + (n_2.n_3) + \dots + (n_b.n_1)}{n^2} < 1 - \frac{1}{b}$$

Which means for the total zero elements in P , or sparsity $S(P)$:

$$1 - \frac{(n-b+1)^2 + (b-1)}{n^2} < S(P) < 1 - \frac{1}{b}$$

Arranging the first terms give us, where we can characterize the sparsity of the $P = S(P)$:

$$\frac{(b-1)(2n-b+2)}{n^2} < S(P) < \frac{b-1}{b}$$

3 Solution to Question 3

Given transition matrix P is symmetric if $P(x, y) = P(y, x)$ for all $x, y \in \mathcal{X}$.

3.1 (a)

Lemma 3.1.1. For a P to have stationary distribution π , with $\mu_0 = \pi$ and $\mu_t = \pi$ for $t \geq 0$, we have $\pi(y) = \sum_{x \in \mathcal{X}} \pi(x)P(x, y)$ for all $y \in \mathcal{X}$. From 1.15 of Levin & Peres.

Since we need to prove for the uniform distribution on $\mathcal{X}$, and assuming $|\mathcal{X}| = n$, we can say that $\pi(x) = \frac{1}{n}$ for $x \in \mathcal{X}$.

From Lemma 3.1.1, by substituting $\pi(x) = \frac{1}{n}$, we get:

$$\pi(y) = \sum_{x \in \mathcal{X}} \pi(x)P(x, y) = \sum_{x \in \mathcal{X}} \frac{1}{n}P(x, y) = \frac{1}{n} \sum_{x \in \mathcal{X}} P(x, y)$$

Since P is symmetric, $P(x, y) = P(y, x)$ for all $x, y \in \mathcal{X}$.

$$\frac{1}{n} \sum_{x \in \mathcal{X}} P(x, y) = \frac{1}{n} \sum_{x \in \mathcal{X}} P(y, x)$$

Since P is a valid transition matrix and stochastic, the sum of probabilities across all outgoing transitions from state y must equal 1 (i.e., $\sum_{x \in \mathcal{X}} P(y, x) = 1$):

$$\frac{1}{n} \sum_{x \in \mathcal{X}} P(y, x) = \frac{1}{n}(1) = \frac{1}{n} = \pi(y)$$

Thus, $\sum_{x \in \mathcal{X}} \pi(x)P(x, y) = \pi(y)$ holds for every state $y \in \mathcal{X}$ from Lemma 3.1.1. The uniform distribution is therefore a stationary distribution of P .

3.2 (b)

From the above section, we prove that for a symmetric P , with $P(x, y) = P(y, x)$ for all $x, y \in \mathcal{X}$, the distribution on $\mathcal{X}$ is uniform with $\pi(x) = \frac{1}{n}$ for $x \in \mathcal{X}$ with $|\mathcal{X}| = n$.

This also means that, $\sum_{x \in \mathcal{X}} P(y, x) = 1$ and $\sum_{y \in \mathcal{X}} P(x, y) = 1$ for both $x, y \in \mathcal{X}$.

The above condition essentially concludes that for symmetric P with uniform distribution π on $\mathcal{X}$, the P matrix is doubly stochastic, meaning both row-sums and column-sums are 1, since x and y can be swapped.

From Lemma 3.1.1 and 1.14 from Levin & Peres, a probability distribution π is a stationary distribution for a transition matrix P if and only if $\pi = \pi P$.

Now, if we can find a counter-example, where a non-symmetric P still has uniform distribution π , satisfying stationary distribution condition $\pi = \pi P$, we can prove the hypothesis.

We assume the P has no self-loops, $P(x, y) \neq P(y, x)$ and $|\mathcal{X}| = 3$.

So, we can form P with $a, b, c > 0$ as:

$$\begin{bmatrix} 0 & a & 1-a \\ b & 0 & 1-b \\ c & 1-c & 0 \end{bmatrix}$$

Since P is non-symmetric, we have

$$a \neq b, 1-a \neq 1-b, 1-c \neq 1-c$$

Since we are dealing with uniform distribution, $\pi = \pi P$, we can write $\pi = (\frac{1}{n}, \frac{1}{n}, \frac{1}{n})$.

Following matrix multiplication of $\pi = \pi P$, from P defined above, we can get:

$$\left(\frac{1}{n}, \frac{1}{n}, \frac{1}{n}\right) = \left(\frac{b+c}{n}, \frac{a+1-c}{n}, \frac{2-a-b}{n}\right)$$

By equality, this gives us:

$$b+c = 1a+1-c = 1 \implies a = c, 2-a-b = 1 \implies a+b = 1$$

If we take, $a = \frac{1}{4}$, then $c = \frac{1}{4}$, and $b = \frac{3}{4}$, making P :

$$\begin{bmatrix} 0 & \frac{1}{4} & \frac{3}{4} \\ \frac{3}{4} & 0 & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} & 0 \end{bmatrix}$$

Hence proven that there exist a non-symmetric P , with uniform distribution π , and stationary distribution satisfied, through example.

4 Solution to Question 4

We have a irreducible and aperiodic markov chain P with finite (assumed) state space $x \in \mathcal{X}$.

We have a reward function $R : \mathcal{X} \rightarrow \mathbb{R}_+$, where $R(x)$ is the reward for being in state $x \in \mathcal{X}$.

Assumption: $R(x)$ is bounded and $|R(x)| \leq R_{max}$ for all $x \in \mathcal{X}$.

4.1 (a)

Given, the average reward value function as for starting from $x = x_0$:

$$V_T(x_0) = \frac{1}{1+T} \mathbf{E} \left[\sum_{t=0}^T R(X_t) | X_0 = x_0 \right]$$

$\pi : \mathcal{X} \rightarrow [0, 1]$ is the unique stationary distribution of markov chain

Considering X as the random variable, with $P^t(x_0, x) = \mathbb{P}(X_t = x | X_0 = x_0)$ being the t -step transition probability for $t \geq 0$, we can write the reward of starting from $X = x_0$ as:

$$R(X_t | X_0 = x_0) = \sum_{x \in \mathcal{X}} P^t(x_0, x) R(x) \quad \forall x \in \mathcal{X}, t \geq 0$$

So the expected reward for $t \in [0, T]$ is:

$$\mathbf{E} \left[\frac{1}{1+T} \sum_{t=0}^T R(X_t | X_0 = x_0) \right] = \frac{1}{1+T} \sum_{t=0}^T \sum_{x \in \mathcal{X}} P^t(x_0, x) R(x) \quad \forall x \in \mathcal{X}$$

So, we can write $V_T(x_0)$ as:

$$V_T(x_0) = \frac{1}{1+T} \sum_{t=0}^T \sum_{x \in \mathcal{X}} P^t(x_0, x) R(x) \quad \forall x \in \mathcal{X}$$

Since the state space $\mathcal{X}$ is finite, we can swap the order of summation:

$$V_T(x_0) = \sum_{x \in \mathcal{X}} R(x) \left(\frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) \right)$$

Because the given Markov chain is irreducible, aperiodic and has π as unique stationary distribution, with a finite state space $\mathcal{X}$, the Definition 1.7, 1.14 from Levin & Peres regarding Markov Chains implies that:

$$\lim_{t \rightarrow \infty} P^t(x_0, x) = \pi(x) \quad \forall x \in \mathcal{X}$$

Lemma 4.1.1. *The series $\sum_{t=0}^{\infty} a_t$ tends to a finite L if the partial sum $s_T = \sum_{t=0}^T a_t \rightarrow L$ for $T \rightarrow \infty$. (Ref to convergent series)*

Lemma 4.1.2. *Based on the Cesaro summation for series and Lemma 4.1.1 (Ref to casero summability series),*

$$\lim_{T \rightarrow \infty} \frac{1}{1+T} \sum_{t=0}^T s_t \rightarrow L$$

Lemma 4.1.3. *Based on the Average of Sequence and Lemma 4.1.1 (Ref to average of sequence),*

$$\lim_{T \rightarrow \infty} \frac{1}{1+T} \sum_{t=0}^T a_t \rightarrow L \quad \text{also} \quad \lim_{T \rightarrow \infty} a_t \rightarrow L$$

If $P^t(x_0, x)$ is considered as a_t

Based on Lemma 4.1.3 and $\lim_{t \rightarrow \infty} P^t(x_0, x) = \pi(x)$, then we can write:

$$\lim_{T \rightarrow \infty} \frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) = \lim_{T \rightarrow \infty} \frac{1}{T+1} \sum_{t=0}^T a_t = \pi(x)$$

Going back to the equation for $V_T(x_0)$ and taking limits to ∞ :

$$\lim_{T \rightarrow \infty} V_T(x_0) = \lim_{T \rightarrow \infty} \sum_{x \in \mathcal{X}} R(x) \left(\frac{1}{T+1} \sum_{t=0}^T P^t(x_0, x) \right)$$

$$\lim_{T \rightarrow \infty} V_T(x_0) = \sum_{x \in \mathcal{X}} R(x) \pi(x)$$

Hence, proved that:

$$V(x_0) = \lim_{T \rightarrow \infty} V_T(x_0) = \sum_{x \in \mathcal{X}} R(x) \pi(x)$$

4.2 (b)

We assume that $\tau_{mix}(\epsilon)$ for the markov chain is known.

There exist a algorithm for $V(x_0)$, with estimator as $\hat{V}_N$, which can run till N simulations.

Given, we run N simulations starting at $x_0 \in \mathcal{X}$ with length $\tau_{mix}(\epsilon)$ where the below is the realization of the n^{th} simulation.:

$$X_n = (X_{n,0}, X_{n,1}, \dots, X_{n,\tau_{mix}(\epsilon)})$$

After setting $Y_n(\epsilon) = X_{n,\tau_{mix}(\epsilon)}$, we get:

$$\hat{V}_N = \frac{1}{N} \sum_{n=1}^N R(Y_n(\epsilon))$$

We have to prove that the Mean Squared Error (MSE) satisfies:

$$\mathbb{E} [(\hat{V}_N - V)^2] \leq R_{\max}^2 \left(\frac{1}{N} + 4\epsilon^2 \right)$$

Now, for any estimator $\hat{V}_N$ targeting a constant parameter V , the MSE decomposes into variance and squared bias (Reference to decomposition here):

$$\mathbb{E} [(\hat{V}_N - V)^2] = \text{Var}(\hat{V}_N) + \left(\mathbb{E}[\hat{V}_N] - V \right)^2$$

Assumming the N simulations are independent realizations, the random variables $Y_n(\epsilon) = X_{n,\tau_{mix}(\epsilon)}$ are independent and identically distributed (i.i.d.).

Therefore,

$$\text{Var}(\hat{V}_N) = \text{Var} \left(\frac{1}{N} \sum_{n=1}^N R(Y_n(\epsilon)) \right) = \frac{1}{N^2} \sum_{n=1}^N \text{Var}(R(Y_n(\epsilon))) = \frac{1}{N} \text{Var}(R(Y_1(\epsilon)))$$

Given, the reward function $R(x)$ is bounded in the range $[-R_{\max}, R_{\max}]$, so $R(Y_1(\epsilon)) \in [a, b]$ where $b - a \leq 2R_{\max}$.

By Popoviciu's inequality on variances, for any bounded random variable $Z \in [a, b]$:

$$\text{Var}(Z) \leq \frac{1}{4}(b - a)^2$$

Applying Popoviciu's inequality to $R(Y_1(\epsilon))$:

$$\text{Var}(R(Y_1(\epsilon))) \leq \frac{1}{4}(R_{\max} - (-R_{\max}))^2 = \frac{1}{4}(2R_{\max})^2 = R_{\max}^2$$

Substituting this back into the variance expression:

$$\text{Var}(\hat{V}_N) \leq \frac{R_{\max}^2}{N}$$

Now by linearity of expectation:

$$\mathbb{E}[\hat{V}_N] = \mathbb{E}[R(Y_1(\epsilon))] = \sum_{x \in \mathcal{X}} P^{\tau_{\text{mix}}(\epsilon)}(x_0, x) R(x)$$

From previous question, $V = \sum_{x \in \mathcal{X}} \pi(x) R(x)$.

Subtracting these two expressions:

$$\mathbb{E}[\hat{V}_N] - V = \sum_{x \in \mathcal{X}} (P^{\tau_{\text{mix}}(\epsilon)}(x_0, x) - \pi(x)) R(x)$$

Taking the absolute value and applying Triangle inequality:

$$|\mathbb{E}[\hat{V}_N] - V| \leq \sum_{x \in \mathcal{X}} |P^{\tau_{\text{mix}}(\epsilon)}(x_0, x) - \pi(x)| |R(x)| \leq R_{\max} \sum_{x \in \mathcal{X}} |P^{\tau_{\text{mix}}(\epsilon)}(x_0, x) - \pi(x)|$$

By definition 4.2 in Levin & Peres, total variation distance is

$$d(t) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P^t(x_0, x) - \pi(x)|$$

From 4.32 in Levin & Peres, at $t = \tau_{\text{mix}}(\epsilon)$, we have $d(\tau_{\text{mix}}(\epsilon)) \leq \epsilon$, which implies:

$$\sum_{x \in \mathcal{X}} |P^{\tau_{\text{mix}}(\epsilon)}(x_0, x) - \pi(x)| \leq 2\epsilon$$

Thus, the bias is bounded by:

$$|\mathbb{E}[\hat{V}_N] - V| \leq 2\epsilon R_{\max}$$

Squaring both sides gives:

$$(\mathbb{E}[\hat{V}_N] - V)^2 \leq 4\epsilon^2 R_{\max}^2$$

Substituting both components into the MSE decomposition back:

$$\mathbb{E}[(\hat{V}_N - V)^2] \leq \frac{R_{\max}^2}{N} + 4\epsilon^2 R_{\max}^2 = R_{\max}^2 \left(\frac{1}{N} + 4\epsilon^2 \right)$$

Hence, proved.

4.3 (c)

We want to find the total number of simulations N and the trajectory length τ such that the estimator $\hat{V}_N$ satisfies:

$$\mathbb{E}[(\hat{V}_N - V)^2] \leq \epsilon$$

From previous question, the MSE bound is given by:

$$\mathbb{E}[(\hat{V}_N - V)^2] \leq R_{\max}^2 \left(\frac{1}{N} + 4\epsilon_0^2 \right)$$

where ϵ_0 is the total variation error associated with running the simulations for length $\tau = \tau_{\text{mix}}(\epsilon_0)$, and e is estimator with respect to MSE.

To ensure the total MSE is bounded by ϵ , we divide ϵ equally between the variance term and the mixing error term, based on Levin & Peres (Eq. 4.34):

Variance Term:

$$\frac{R_{\text{max}}^2}{N} \leq \frac{\epsilon}{2} \implies N \geq \frac{2R_{\text{max}}^2}{\epsilon}$$

Mixing Error Term:

$$4R_{\text{max}}^2 \epsilon_0^2 \leq \frac{\epsilon}{2} \implies \epsilon_0^2 \leq \frac{\epsilon}{8R_{\text{max}}^2} \implies \epsilon_0 \leq \sqrt{\frac{\epsilon}{8R_{\text{max}}^2}}$$

Using the hint from Levin & Peres (Eq. 4.36):

$$\tau_{\text{mix}}(\epsilon_0) \leq \lceil \log_2(\epsilon_0^{-1}) \rceil \tau_{\text{mix}}\left(\frac{1}{4}\right)$$

Substitute $\epsilon_0 = \sqrt{\frac{\epsilon}{8R_{\text{max}}^2}}$ into ϵ_0^{-1} :

$$\epsilon_0^{-1} = \sqrt{\frac{8R_{\text{max}}^2}{\epsilon}} = \left(\frac{8R_{\text{max}}^2}{\epsilon}\right)^{1/2}$$

Taking the base-2 logarithm:

$$\log_2(\epsilon_0^{-1}) = \log_2\left(\left(\frac{8R_{\text{max}}^2}{\epsilon}\right)^{1/2}\right) = \frac{1}{2} \log_2\left(\frac{8R_{\text{max}}^2}{\epsilon}\right)$$

Therefore, each trajectory length τ needs to be set to:

$$\tau = \tau_{\text{mix}}(\epsilon_0) \leq \left\lceil \frac{1}{2} \log_2\left(\frac{8R_{\text{max}}^2}{\epsilon}\right) \right\rceil \tau_{\text{mix}}\left(\frac{1}{4}\right)$$

The total computational samples required is the product of the number of simulations N and the trajectory length τ per simulation:

$$\text{Total Samples} = N \times \tau$$

Substituting $N = \left\lceil \frac{2R_{\text{max}}^2}{\epsilon} \right\rceil$ and τ :

$$\text{Total Samples} = \left\lceil \frac{2R_{\text{max}}^2}{\epsilon} \right\rceil \cdot \left\lceil \frac{1}{2} \log_2\left(\frac{8R_{\text{max}}^2}{\epsilon}\right) \right\rceil \tau_{\text{mix}}\left(\frac{1}{4}\right)$$

Hence proved.

Note: The final term is slightly different from the question, due to $\frac{\epsilon}{2}$ bound based on Levin & Peres 4.34.

$$\text{Total Samples} = \left\lceil \frac{2R_{\text{max}}^2}{\epsilon} \right\rceil \cdot \left\lceil \frac{1}{2} \log_2\left(\frac{8R_{\text{max}}^2}{\epsilon}\right) \right\rceil \tau_{\text{mix}}\left(\frac{1}{4}\right)$$

5 Solution to Question 5

Given, $T : \mathcal{X} \rightarrow \mathcal{X}$ be an α -contraction in some norm $\|\cdot\|$.

Lemma 5.0.1. *Let $T : \mathcal{X} \rightarrow \mathcal{X}$ be a mapping (operator). T is an α -contraction with respect to the norm $\|\cdot\|$ if there exists a constant $\alpha \in [0, 1)$ such that for all $x, y \in \mathcal{X}$: $\|T(x) - T(y)\| \leq \alpha\|x - y\|$. α is called the contraction constant. Geometrically, T strictly shrinks the distance between any two points by a factor of at least α . Reference to lecture note*

5.1 (a)

Using Lemma 5.0.1, suppose there exist two distinct fixed points $x^*, y^* \in \mathcal{X}$ such that $x^* \neq y^*$.

By definition of a fixed point:

$$T(x^*) = x^* \quad \text{and} \quad T(y^*) = y^*$$

The norm distance between x^* and y^* :

$$\|x^* - y^*\| = \|T(x^*) - T(y^*)\|$$

Applying the contraction inequality:

$$\|x^* - y^*\| \leq a\|x^* - y^*\|$$

Subtracting $a\|x^* - y^*\|$ from both sides:

$$(1 - a)\|x^* - y^*\| \leq 0 \tag{5.1.1}$$

Now, since $x^* \neq y^*$, the distance between them is strictly positive, $\|x^* - y^*\| > 0$.

Also, since T is an a -contraction, $a \in [0, 1)$, which means $(1 - a) > 0$. Multiplying two strictly positive real numbers yields a strictly positive result:

$$(1 - a)\|x^* - y^*\| > 0 \tag{5.1.2}$$

As we can see, Equation 5.1.1 and Equation 5.1.2 contradict with each other.

Hence, the assumption that $x^* \neq y^*$ must be false. Therefore, $x^* = y^*$, proving that the fixed point x^* is unique.

5.2 (b)

We can consider $\mathcal{X} = (0, 1)$, with the standard Euclidean metric $d(x, y) = |x - y|$.

Lets $T : \mathcal{X} \rightarrow \mathcal{X}$ by:

$$T(x) = \frac{x}{k} \quad \text{for some } k > 0$$

For any $x, y \in (0, 1)$:

$$d(T(x), T(y)) = \left| \frac{x}{k} - \frac{y}{k} \right| = \frac{1}{k} |x - y| = \frac{1}{k} d(x, y)$$

Since $\alpha = \frac{1}{k} < 1$, the function T is an α -contraction on $\mathcal{X}$.

A fixed point $x^* \in \mathcal{X}$ must satisfy $T(x^*) = x^*$:

$$\frac{x^*}{k} = x^* \implies x^* = 0$$

However, $0 \notin \mathcal{X}$ because $\mathcal{X} = (0, 1)$ is an open interval that excludes 0. Therefore, there exists no fixed point within the set $\mathcal{X}$ for the given $T(x) = \frac{x}{k}$ with α -contraction on $\mathcal{X}$

5.3 (c)

Lets assume an arbitrary starting point $x_0 \in \mathcal{X}$. We then assume a sequence $\{x_k\}_{k=0}^{\infty}$ by repeatedly applying T :

$$x_k = T(x_{k-1}) = T^k(x_0) \quad \text{for } k \geq 1$$

Since T is an α -contraction, for any $x, y \in \mathcal{X}$, we have:

$$\|T(x) - T(y)\| \leq \alpha \|x - y\|$$

Applying this recursively to successive terms in our sequence gives:

$$\|x_{k+1} - x_k\| = \|T(x_k) - T(x_{k-1})\| \leq \alpha \|x_k - x_{k-1}\| \leq \alpha^2 \|x_{k-1} - x_{k-2}\| \leq \dots \leq \alpha^k \|x_1 - x_0\|$$

For any integers $m > n \geq 0$, use the triangle inequality to expand $\|x_m - x_n\|$:

$$\|x_m - x_n\| \leq \sum_{i=n}^{m-1} \|x_{i+1} - x_i\|$$

Substitute the upper bound for each step derived above:

$$\|x_m - x_n\| \leq \sum_{i=n}^{m-1} \alpha^i \|x_1 - x_0\| = \alpha^n \|x_1 - x_0\| \sum_{j=0}^{m-n-1} \alpha^j$$

Summing the geometric series gives us:

$$\sum_{j=0}^{m-n-1} \alpha^j \leq \sum_{j=0}^{\infty} \alpha^j = \frac{1}{1 - \alpha}$$

Thus:

$$\|x_m - x_n\| \leq \frac{\alpha^n}{1 - \alpha} \|x_1 - x_0\|$$

Since $\alpha \in (0, 1)$, as $n \rightarrow \infty$, $\alpha^n \rightarrow 0$. Therefore, for any $\epsilon > 0$, we can choose a sufficiently large integer N such that for all $m > n > N$:

$$\|x_m - x_n\| < \epsilon$$

This proves that $\{x_k\}$ is a Cauchy sequence in $\mathcal{X}$, based on the notes given for the Cauchy sequence in the homework and also from this reference.

Because $\mathcal{X}$ is a complete metric space, every Cauchy sequence in $\mathcal{X}$ converges to a limit inside $\mathcal{X}$. Let $x^* \in \mathcal{X}$ be the limit of the sequence:

$$\lim_{k \rightarrow \infty} x_k = x^*$$

Now for $T(x^*) = x^*$, let's consider the distance $\|T(x^*) - x^*\|$:

$$\|T(x^*) - x^*\| \leq \|T(x^*) - x_{k+1}\| + \|x_{k+1} - x^*\| \quad (\text{by triangle inequality})$$

Since $x_{k+1} = T(x_k)$, we can write the first term using the contraction property:

$$\|T(x^*) - T(x_k)\| \leq \alpha \|x^* - x_k\|$$

Thus:

$$\|T(x^*) - x^*\| \leq \alpha \|x^* - x_k\| + \|x_{k+1} - x^*\|$$

Taking the limit as $k \rightarrow \infty$, we get $\|x^* - x_k\| \rightarrow 0$ and $\|x_{k+1} - x^*\| \rightarrow 0$, by Lemma 4.1.3.

This makes $\|T(x^*) - x^*\| = 0$, which implies:

$$T(x^*) = x^*$$

Hence, $x^* \in \mathcal{X}$ is a fixed point of $T(\cdot)$. Combined with the uniqueness proved earlier, x^* is the unique fixed point in $\mathcal{X}$.

6 Solution to Question 6

6.1 (a)

The given operator or matrix $W \in \mathbb{R}^{N \times N}$ is a contraction with respect to a norm $\|\cdot\|$ if there exists a constant $\alpha < 1$ such that for all $x, y \in \mathbb{R}^N$:

$$\|Wx - Wy\| \leq \alpha \|x - y\|$$

By linearity, setting $z = x - y$, this is equivalent to:

$$\|Wz\| \leq \alpha \|z\| \quad \text{for all } z \in \mathbb{R}^N$$

This implies that the induced matrix norm obeys (Reference to Lecture Notes 1)(Reference to Lecture Notes 2):

$$\|W\| = \max_{z \neq 0} \frac{\|Wz\|}{\|z\|} \leq \alpha < 1$$

Consider the vector $z = \mathbf{1}$. Applying W to z :

$$W\mathbf{1} = \mathbf{1}$$

Taking the norm on both sides for any valid norm $\|\cdot\|$:

$$\|W\mathbf{1}\| = \|\mathbf{1}\|$$

Suppose, for the sake of contradiction, that W is a contraction in some norm $\|\cdot\|$ with contraction factor $\alpha < 1$.

Then, for $z = \mathbf{1} \neq \mathbf{0}$, we must have:

$$\|W\mathbf{1}\| \leq \alpha \|\mathbf{1}\|$$

Substituting $\|W\mathbf{1}\| = \|\mathbf{1}\|$ into the inequality:

$$\|\mathbf{1}\| \leq \alpha \|\mathbf{1}\|$$

Since $\mathbf{1} \neq \mathbf{0}$, its norm is strictly positive ($\|\mathbf{1}\| > 0$).

Dividing both sides by $\|\mathbf{1}\|$ yields:

$$1 \leq \alpha$$

This directly contradicts the requirement that $\alpha < 1$.

Because W preserves the norm of the all-ones vector $\mathbf{1}$ under any matrix/vector norm, no constant $\gamma < 1$ can exist such that $\|Wx\| \leq \gamma\|x\|$ for all $x \in \mathbb{R}^N$. Therefore, W cannot be a contraction in any norm.

6.2 (b)

To show the set of fixed points of the algorithm $x(k+1) = Wx(k)$ is $\text{span}(\mathbf{1}) = \{a\mathbf{1} \in \mathbb{R}^N \mid a \in \mathbb{R}\}$, we need to prove both inclusions of the set equality:

$$\mathcal{S} = \{x \in \mathbb{R}^N \mid Wx = x\} = \text{span}(\mathbf{1})$$

A vector $x^* \in \mathbb{R}^N$ is a fixed point of the update rule if and only if:

$$Wx^* = x^*$$

This is equivalent to stating that x^* is a right eigenvector of W associated with the eigenvalue $\lambda = 1$.

Therefore, the set of fixed points is precisely the eigen-space corresponding to $\lambda = 1$:

$$\text{Eig}(W, 1) = \{x \in \mathbb{R}^N \mid Wx = 1 \cdot x\}$$

By definition, W is a doubly stochastic matrix. Therefore, each row sum of W equals 1, which gives:

$$W\mathbf{1} = \mathbf{1}$$

For any scalar $a \in \mathbb{R}$, by linearity of matrix multiplication:

$$W(a\mathbf{1}) = a(W\mathbf{1}) = a\mathbf{1}$$

Thus, any vector of the form $a\mathbf{1}$ is a fixed point, showing that:

$$\text{span}(\mathbf{1}) \subseteq \{x \in \mathbb{R}^N \mid Wx = x\}$$

To establish that there are no other linearly independent fixed points, we apply the Perron-Frobenius Theorem for non-negative matrices.

Regarding the properties of W , $W_{nm} \geq 0$ for all $n, m \in \{1, \dots, N\}$.

Since W inherits the sparsity structure of the underlying communication graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$ and $\mathcal{G}$ is connected, W is an irreducible matrix.

Because W is doubly stochastic, its spectral radius is $\rho(W) = 1$.

According to the Perron-Frobenius theorem, for an irreducible, non-negative matrix W with spectral radius $r = \rho(W) = 1$,

the spectral radius $r = 1$ is a simple eigenvalue of W (algebraic multiplicity = 1) and the associated eigenspace $\text{Eig}(W, 1)$ is one-dimensional (geometric multiplicity = 1).

Since $\dim(\text{Eig}(W, 1)) = 1$ and we already found a non-zero element $\mathbf{1} \in \text{Eig}(W, 1)$, $\mathbf{1}$ must span the entire eigenspace:

$$\text{Eig}(W, 1) = \text{span}(\mathbf{1})$$

Combining both inclusions, the set of fixed points for the algorithm $x(k+1) = Wx(k)$ is uniquely given by:

$$\{x \in \mathbb{R}^N \mid Wx = x\} = \text{span}(\mathbf{1}) = \{a\mathbf{1} \in \mathbb{R}^N \mid a \in \mathbb{R}\}$$

6.3 (c)

The set $\mathcal{A}(\bar{x})$ is defined as:

$$\mathcal{A}(\bar{x}) = \left\{ x \in \mathbb{R}^N \mid \frac{1}{N} \sum_{n=1}^N x_n = \bar{x} \right\}$$

Using vector notation with $\mathbf{1} = [1, 1, \dots, 1]^T \in \mathbb{R}^N$, the condition $\frac{1}{N} \sum_{n=1}^N x_n = \bar{x}$ can be written compactly as:

$$\frac{1}{N} \mathbf{1}^T x = \bar{x}$$

We assume $x \in \mathcal{A}(\bar{x})$, which gives us the hypothesis:

$$\frac{1}{N} \mathbf{1}^T x = \bar{x}$$

We want to prove that $Wx \in \mathcal{A}(\bar{x})$, which means we must show that:

$$\frac{1}{N} \mathbf{1}^T (Wx) = \bar{x}$$

For evaluating the term $\frac{1}{N}\mathbf{1}^T(Wx)$, we use Matrix Associativity:

$$\frac{1}{N}\mathbf{1}^T(Wx) = \frac{1}{N}(\mathbf{1}^TW)x$$

The matrix W is doubly stochastic. By definition, every column of W sums to 1, which gives the property:

$$\mathbf{1}^TW = \mathbf{1}^T$$

Substituting $\mathbf{1}^TW = \mathbf{1}^T$ into the Expression:

$$\frac{1}{N}(\mathbf{1}^TW)x = \frac{1}{N}\mathbf{1}^Tx$$

Since $x \in \mathcal{A}(\bar{x})$, we know that $\frac{1}{N}\mathbf{1}^Tx = \bar{x}$.

Therefore:

$$\frac{1}{N}\mathbf{1}^T(Wx) = \bar{x}$$

This satisfies the condition for Wx to be in $\mathcal{A}(\bar{x})$.

$$\implies Wx \in \mathcal{A}(\bar{x})$$

6.4 (d)

The definition of the vector of ones $\mathbf{1} = [1, 1, \dots, 1]^T \in \mathbb{R}^N$ and the set:

$$\mathcal{A}(\bar{x}) = \left\{ x \in \mathbb{R}^N \left| \frac{1}{N} \sum_{n=1}^N x_n = \bar{x} \right. \right\}$$

For any $x \in \mathcal{A}(\bar{x})$, the sum of its elements satisfies $\mathbf{1}^Tx = \sum_{n=1}^N x_n = N\bar{x}$. Multiplying both sides by $\frac{1}{N}\mathbf{1}$:

$$\frac{1}{N}\mathbf{1}\mathbf{1}^Tx = \frac{1}{N}\mathbf{1}(N\bar{x}) = \bar{x}\mathbf{1}$$

Using the identity $\frac{1}{N}\mathbf{1}\mathbf{1}^Tx = \bar{x}\mathbf{1}$ derived above, we can rewrite $Wx - \bar{x}\mathbf{1}$ by factoring out x :

$$Wx - \bar{x}\mathbf{1} = Wx - \frac{1}{N}\mathbf{1}\mathbf{1}^Tx = \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T \right) x \quad (6.4.1)$$

Since W is doubly stochastic, $W\mathbf{1} = \mathbf{1}$:

$$W(\bar{x}\mathbf{1}) = \bar{x}W\mathbf{1} = \bar{x}\mathbf{1}$$

Using $\mathbf{1}^T\mathbf{1} = N$:

$$\left(\frac{1}{N}\mathbf{1}\mathbf{1}^T \right) (\bar{x}\mathbf{1}) = \frac{\bar{x}}{N}\mathbf{1}(\mathbf{1}^T\mathbf{1}) = \frac{\bar{x}}{N}\mathbf{1}(N) = \bar{x}\mathbf{1}$$

So,

$$\left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)(\bar{x}\mathbf{1}) = \bar{x}\mathbf{1} - \bar{x}\mathbf{1} = \mathbf{0}$$

We can write like Equation 6.4.1 below since above $\left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)(\bar{x}\mathbf{1}) = \mathbf{0}$:

$$Wx - \bar{x}\mathbf{1} = \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)x - \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)(\bar{x}\mathbf{1}) = \left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)(x - \bar{x}\mathbf{1})$$

Taking the ℓ_2 -norm of both sides above and applying the sub-multiplicative property of matrix norms (Reference to Lecture Notes) $\|Ay\|_2 \leq \|A\|_2\|y\|_2$:

$$\|Wx - \bar{x}\mathbf{1}\|_2 = \left\|\left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)(x - \bar{x}\mathbf{1})\right\|_2 \leq \left\|W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right\|_2 \|x - \bar{x}\mathbf{1}\|_2$$

Setting $\sigma = \left\|W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right\|_2$ gives:

$$\|Wx - \bar{x}\mathbf{1}\|_2 \leq \sigma \|x - \bar{x}\mathbf{1}\|_2$$

Lemma 6.4.1. *The operator norm (or spectral norm) $\|A\|_2$ of a real matrix A is equal to its largest singular value, which is the square root of the largest eigenvalue of $A^T A$ (or simply the maximum absolute value eigenvalue if the matrix is symmetric/normal).*

Let the matrix $M = W - \frac{1}{N}\mathbf{1}\mathbf{1}^T$.

The spectral radius of W is $\rho(W) = 1$, and its primary eigenvector corresponding to $\lambda_1 = 1$ is $\mathbf{1}$ (since $W\mathbf{1} = \mathbf{1}$).

The matrix $\frac{1}{N}\mathbf{1}\mathbf{1}^T$ is a projection matrix onto $\text{span}(\mathbf{1})$ with eigenvalue 1 for eigenvector $\mathbf{1}$ and 0 for all vectors orthogonal to $\mathbf{1}$.

Subtracting $\frac{1}{N}\mathbf{1}\mathbf{1}^T$ from W effectively shifts the top eigenvalue corresponding to $\mathbf{1}$ from 1 to 0, while leaving all other eigenvalues (and their corresponding invariant subspaces orthogonal to $\mathbf{1}$) unchanged.

Thus, the eigenvalues of $M = W - \frac{1}{N}\mathbf{1}\mathbf{1}^T$ are $\{0, \lambda_2, \lambda_3, \dots, \lambda_N\}$, where λ_i are the remaining eigenvalues of W .

Because W is a non-negative, irreducible, and aperiodic (since $W_{nn} > 0$, self-loops) doubly stochastic matrix, by the Perron-Frobenius theorem:

- $\lambda_1 = 1$ is a simple eigenvalue.
- All other eigenvalues satisfy $|\lambda_i| < 1$ for all $i \geq 2$.

Therefore:

$$\sigma = \left\|W - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right\|_2 = \max_{i \geq 2} |\lambda_i| < 1$$

Hence,

$$\|Wx - \bar{x}\mathbf{1}\|_2 \leq \sigma \|x - \bar{x}\mathbf{1}\|_2 \quad \text{with } \sigma < 1$$

6.5 (e)

Since $W \in \mathbb{R}^{N \times N}$ is symmetric, by the Spectral Theorem, W can be orthogonally diagonalized.

It has N real eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_N$ and an orthonormal basis of eigenvectors $\{v_1, v_2, \dots, v_N\}$.

Since W is doubly stochastic, $W\mathbf{1} = \mathbf{1}$.

Thus, $\mathbf{1}$ is an eigenvector associated with the eigenvalue 1.

Normalizing this vector gives the unit eigenvector:

$$v_1 = \frac{1}{\sqrt{N}}\mathbf{1}$$

By the Perron-Frobenius theorem for irreducible non-negative matrices, the spectral radius of W is $\rho(W) = 1$, and $\lambda_1 = 1$ is a simple eigenvalue.

We can order the eigenvalues in decreasing order of their absolute values:

$$1 = \lambda_1 > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_N|$$

where λ_2 is the second largest eigenvalue in absolute value.

Since $\{v_1, v_2, \dots, v_N\}$ forms an orthonormal basis for $\mathbb{R}^N$, we can express the initial state vector $x(0)$ as a linear combination of these eigenvectors:

$$x(0) = c_1 v_1 + \sum_{i=2}^N c_i v_i$$

To find c_1 , we can compute the inner product of $x(0)$ with v_1 :

$$c_1 = v_1^T x(0) = \left(\frac{1}{\sqrt{N}}\mathbf{1} \right)^T x(0) = \frac{1}{\sqrt{N}} \sum_{n=1}^N x_n(0) = \frac{N\bar{x}}{\sqrt{N}} = \sqrt{N}\bar{x}$$

Substituting c_1 back into the first term of $x(0)$:

$$c_1 v_1 = (\sqrt{N}\bar{x}) \left(\frac{1}{\sqrt{N}}\mathbf{1} \right) = \mathbf{1}\bar{x}$$

Thus, the initial vector decomposes into the average vector and a component in the orthogonal subspace:

$$x(0) = \mathbf{1}\bar{x} + \sum_{i=2}^N c_i v_i$$

Since $x(k) = W^k x(0)$, applying W^k to the decomposition gives:

$$x(k) = W^k \left(\mathbf{1}\bar{x} + \sum_{i=2}^N c_i v_i \right)$$

Since $W^k \mathbf{1} = \mathbf{1}$ and $W^k v_i = \lambda_i^k v_i$:

$$x(k) = \mathbf{1}\bar{x} + \sum_{i=2}^N c_i \lambda_i^k v_i$$

Subtracting the average vector $\mathbf{1}\bar{x}$ from both sides gives the error vector:

$$x(k) - \mathbf{1}\bar{x} = \sum_{i=2}^N c_i \lambda_i^k v_i$$

Taking the norm $\|\cdot\|_2$ of both sides and using the fact that $\{v_2, \dots, v_N\}$ are orthonormal vectors:

$$\|x(k) - \mathbf{1}\bar{x}\|_2 = \left\| \sum_{i=2}^N c_i \lambda_i^k v_i \right\|_2 = \sqrt{\sum_{i=2}^N |c_i|^2 |\lambda_i|^{2k}}$$

Since $|\lambda_i| \leq |\lambda_2|$ for all $i \geq 2$:

$$|\lambda_i|^{2k} \leq |\lambda_2|^{2k} \quad \forall i = 2, \dots, N$$

Factoring out $|\lambda_2|^k$:

$$\|x(k) - \mathbf{1}\bar{x}\|_2 \leq \sqrt{\sum_{i=2}^N |c_i|^2 |\lambda_2|^{2k}} = \left(\sqrt{\sum_{i=2}^N |c_i|^2} \right) |\lambda_2|^k$$

Using the constant $A = \sqrt{\sum_{i=2}^N |c_i|^2}$, we get at the final bound:

$$\|x(k) - \mathbf{1}\bar{x}\|_2 \leq A |\lambda_2|^k \quad \forall k$$

Hence, proved.

7 Solution to Question 7

Let x^* be the unique minimizer of $f(x)$.

By the assumptions on optimality conditions and first-order optimization, we have:

$$\nabla f(x^*) = 0$$

The noisy gradient descent update rule is given by:

$$x(k+1) = x(k) - \gamma (\nabla f(x(k)) + e(k))$$

Subtracting x^* from both sides:

$$x(k+1) - x^* = (x(k) - x^*) - \gamma \nabla f(x(k)) - \gamma e(k)$$

Taking the norm square $\|\cdot\|^2$ on both sides:

$$\|x(k+1) - x^*\|^2 = \|((x(k) - x^*) - \gamma \nabla f(x(k))) - \gamma e(k)\|^2$$

Using the identity $\|a - b\|^2 = \|a\|^2 - 2\langle a, b \rangle + \|b\|^2$ with $a = (x(k) - x^*) - \gamma \nabla f(x(k))$ and $b = \gamma e(k)$:

$$\begin{aligned} \|x(k+1) - x^*\|^2 &= \|x(k) - x^* - \gamma \nabla f(x(k))\|^2 \\ &\quad - 2\gamma \langle x(k) - x^* - \gamma \nabla f(x(k)), e(k) \rangle \\ &\quad + \gamma^2 \|e(k)\|^2 \end{aligned}$$

Expanding the first term $\|x(k) - x^* - \gamma \nabla f(x(k))\|^2$:

$$\|x(k) - x^* - \gamma \nabla f(x(k))\|^2 = \|x(k) - x^*\|^2 - 2\gamma \langle \nabla f(x(k)), x(k) - x^* \rangle + \gamma^2 \|\nabla f(x(k))\|^2$$

We take the expectation with respect to the up to step k ($\mathcal{F}_k$).

Since $e(k)$ is IID zero-mean noise independent of $x(k)$, we have:

$$\mathbb{E}[e(k) \mid \mathcal{F}_k] = 0 \quad \text{and} \quad \mathbb{E}[\|e(k)\|^2 \mid \mathcal{F}_k] = \sigma^2$$

Taking expectations gives:

$$\mathbb{E}[\|x(k+1) - x^*\|^2 \mid \mathcal{F}_k] = \|x(k) - x^*\|^2 - 2\gamma \langle \nabla f(x(k)), x(k) - x^* \rangle + \gamma^2 \|\nabla f(x(k))\|^2 + \gamma^2 \sigma^2$$

Since f is L -smooth and μ -convex, for any x, y :

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{L\mu}{L + \mu} \|x - y\|^2 + \frac{1}{\mu + L} \|\nabla f(x) - \nabla f(y)\|^2$$

Setting $x = x(k)$ and $y = x^*$, and using $\nabla f(x^*) = 0$:

$$\langle \nabla f(x(k)), x(k) - x^* \rangle \geq \frac{L\mu}{L + \mu} \|x(k) - x^*\|^2 + \frac{1}{\mu + L} \|\nabla f(x(k))\|^2$$

Multiplying by -2γ and changing the inequality:

$$-2\gamma \langle \nabla f(x(k)), x(k) - x^* \rangle \leq -\frac{2\gamma L\mu}{L + \mu} \|x(k) - x^*\|^2 - \frac{2\gamma}{\mu + L} \|\nabla f(x(k))\|^2$$

Substituting this back into expectation equation:

$$\begin{aligned} \mathbb{E}[\|x(k+1) - x^*\|^2 \mid \mathcal{F}_k] &\leq \left(1 - \frac{2\gamma L\mu}{L + \mu}\right) \|x(k) - x^*\|^2 \\ &\quad - \gamma \left(\frac{2}{\mu + L} - \gamma\right) \|\nabla f(x(k))\|^2 + \gamma^2 \sigma^2 \end{aligned}$$

Taking $\gamma \in \left(0, \frac{2}{L + \mu}\right)$, we have:

$$\frac{2}{\mu + L} - \gamma > 0$$

Since $-\gamma \left(\frac{2}{\mu+L} - \gamma \right) \|\nabla f(x(k))\|^2 \leq 0$, we can drop the second term to get an upper bound:

$$\mathbb{E} [\|x(k+1) - x^*\|^2 \mid \mathcal{F}_k] \leq \left(1 - \frac{2\gamma L\mu}{L+\mu} \right) \|x(k) - x^*\|^2 + \gamma^2 \sigma^2$$

Taking total expectation on both sides:

$$\mathbb{E} [\|x(k+1) - x^*\|^2] \leq \left(1 - \frac{2\gamma L\mu}{L+\mu} \right) \mathbb{E} [\|x(k) - x^*\|^2] + \gamma^2 \sigma^2$$

Assuming $c = 1 - \frac{2\gamma L\mu}{L+\mu}$.

Applying the recursion iteratively from $k = 0$ to step k :

$$\mathbb{E} \|x(k+1) - x^*\|^2 \leq c^{k+1} \mathbb{E} \|x(0) - x^*\|^2 + \gamma^2 \sigma^2 \sum_{i=0}^k c^i$$

Using the geometric series bound $\sum_{i=0}^k c^i \leq \sum_{i=0}^{\infty} c^i = \frac{1}{1-c}$:

$$\frac{1}{1-c} = \frac{1}{\frac{2\gamma L\mu}{L+\mu}} = \frac{L+\mu}{2\gamma L\mu}$$

Substituting this in:

$$\gamma^2 \sigma^2 \left(\frac{L+\mu}{2\gamma L\mu} \right) = \gamma \sigma^2 \frac{\mu+L}{2\mu L}$$

Hence, we arrive at the bound:

$$\mathbb{E} \|x(k+1) - x^*\|^2 \leq \left(1 - \frac{2\gamma L\mu}{\mu+L} \right)^{k+1} \mathbb{E} \|x(0) - x^*\|^2 + \gamma \sigma^2 \frac{\mu+L}{2\mu L}$$

Now, as $k \rightarrow \infty$, the first term decays to zero:

$$\lim_{k \rightarrow \infty} \mathbb{E} \|x(k+1) - x^*\|^2 \leq \gamma \sigma^2 \frac{\mu+L}{2\mu L}$$

Since the asymptotic error bound is $\propto \gamma$, by choosing a smaller step size γ , the asymptotic steady-state error $(\gamma \sigma^2 \frac{\mu+L}{2\mu L})$ can be made arbitrarily small.

As $\gamma \rightarrow 0$, the contraction factor $c = 1 - \frac{2\gamma L\mu}{\mu+L}$ gets closer to 1, the rate of decay $(1 - \frac{2\gamma L\mu}{\mu+L})^{k+1}$ becomes extremely slow. Thus, to reach a high accuracy level, a significantly larger number of iterations k is required.

8 Solution to Question 8

We have two definitions given.

Assuming we have uniform probability distribution, it is also given that a sequence of random variables X_k defined on the sample space $\Omega = [0, 1]$, $X_k \xrightarrow{\mathbb{P}} 0$ (converges in probability to 0).

And at the same time, $P(\lim_{k \rightarrow \infty} X_k = 0) \neq 1$ (does not converge almost surely to 0).

Let the sample space be $\Omega = [0, 1]$ equipped with the uniform distribution P , so the probability of an interval $[a, b] \subseteq [0, 1]$ is simply its length $b - a$.

We create intervals I_k that shrink over time but sweep back and forth across $[0, 1]$ infinitely many times,

$$\begin{aligned} I_1 &= [0, 1] \\ I_2 &= [0, 1/2], \quad I_3 = [1/2, 1] \\ I_4 &= [0, 1/4], \quad I_5 = [1/4, 2/4], \quad I_6 = [2/4, 3/4], \quad I_7 = [3/4, 1] \dots \end{aligned}$$

For a general step $n \geq 1$ and index $m \in \{0, 1, \dots, 2^n - 1\}$, we set the iteration index $k = 2^n + m$ and define the interval:

$$I_k = \left[\frac{m}{2^n}, \frac{m+1}{2^n} \right]$$

Now, define the random variable $X_k(\omega)$ as an indicator function:

$$X_k(\omega) = \begin{cases} 1 & \text{if } \omega \in I_k \\ 0 & \text{if } \omega \notin I_k \end{cases}$$

By choosing any $\epsilon \in (0, 1]$. The event $|X_k - 0| \geq \epsilon$ happens if and only if $X_k = 1$.

By definition of X_k , $X_k = 1$ occurs when $\omega \in I_k$. The probability of this event is simply the length of I_k :

$$P(|X_k - 0| \geq \epsilon) = P(X_k = 1) = P(\omega \in I_k) = \text{length}(I_k) = \frac{1}{2^n}$$

As $k \rightarrow \infty$, $n \rightarrow \infty$, so:

$$\lim_{k \rightarrow \infty} P(|X_k - 0| \geq \epsilon) = \lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$$

Hence, X_k Converges in Probability to 0.

Now, if we pick any point $\omega \in [0, 1]$, then for every block size 2^n , the intervals I_k partition the entire interval $[0, 1]$. Thus, ω must fall into at least one interval I_k for each n .

As $k \rightarrow \infty$, the sequence of numbers $X_1(\omega), X_2(\omega), X_3(\omega), \dots$ will contain infinitely many 1's (whenever $\omega \in I_k$) and infinitely many 0's (whenever $\omega \notin I_k$).

Because the sequence $X_k(\omega)$ infinitely alternates between 0 and 1, the limit $\lim_{k \rightarrow \infty} X_k(\omega)$ does not exist for any $\omega \in [0, 1]$.

Since $\lim_{k \rightarrow \infty} X_k(\omega)$ fails to exist to a constant for all $\omega \in [0, 1]$, hence it does not converge almost surely to 0 due to indicator function's behavior of alternating value.

9 Solution to Question 9

There are two inequalities given. One is Markov's inequality and second is Chebychev's inequality.

9.1 (a)

We want to prove that the probability $P(|X - \mu| \geq \alpha) \leq \frac{\text{Var } X}{\alpha^2}$.

We can say that:

$$P(|X - \mu| \geq \alpha) = P(|X - \mu|^2 \geq \alpha^2)$$

Let $Z = (X - \mu)^2$.

We get $Z \geq 0$ (since any squared real number is non-negative).

Z is a random variable.

Thus, the event becomes:

$$P(Z \geq \alpha^2)$$

Since $Z \geq 0$ and $\alpha^2 > 0$, we can use given Markov's inequality to the random variable Z with the threshold parameter α^2 :

$$P(Z \geq \alpha^2) \leq \frac{\mathbb{E}[Z]}{\alpha^2}$$

Substituting $Z = (X - \mu)^2$ back into the expression:

$$P((X - \mu)^2 \geq \alpha^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{\alpha^2}$$

By definition, the variance of X is the expected value of the squared deviation from its mean:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

So,

$$P(Z \geq \alpha^2) \leq \frac{\mathbb{E}[Z]}{\alpha^2} \implies P(|X - \mu| \geq \alpha) \leq \frac{\text{Var}(X)}{\alpha^2}$$

Hence, Chebychev's inequality is proven here.

9.2 (b)

Using the random variable given, we have series $S_n = \frac{1}{n} \sum_{i=1}^n X_i$ and the linearity of expectation, since each X_i has mean μ :

$$\mathbb{E}[S_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \cdot (n\mu) = \mu$$

Using the properties of variance for independent random variables (since $X_1, X_2, \dots, X_n$ are IID with variance σ^2), we can say that:

$$\text{Var}(S_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot (n\sigma^2) = \frac{\sigma^2}{n}$$

Now, using Chebyshev's inequality for a random variable Y with mean μ_Y and any $\alpha > 0$:

$$P(|Y - \mu_Y| \geq \alpha) \leq \frac{\text{Var}(Y)}{\alpha^2}$$

Substituting $Y = S_n$, where $\mu_Y = \mathbb{E}[S_n] = \mu$ and $\text{Var}(S_n) = \frac{\sigma^2}{n}$:

$$P(|S_n - \mu| \geq \alpha) \leq \frac{\frac{\sigma^2}{n}}{\alpha^2} = \frac{\sigma^2}{\alpha^2 n}$$

Replacing α as ϵ , we prove:

$$P(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2 n}$$

Now, taking the limit as $n \rightarrow \infty$ on both sides:

$$\lim_{n \rightarrow \infty} P(|S_n - \mu| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{\sigma^2}{\epsilon^2 n} = 0$$

Since probabilities are non-negative, this implies:

$$\lim_{n \rightarrow \infty} P(|S_n - \mu| \geq \epsilon) = 0$$

By definition of Convergence of Probability from previous questions, we can conclude that the above S_n converges in probability to μ for a $\epsilon < 0$.

9.3 (c)

We want to ensure that the probability of the estimation error being at most ϵ is at least $1 - \delta$:

$$P(|S_n - \mu| \leq \epsilon) \geq 1 - \delta$$

Using the complement rule for probabilities, this condition is equivalent to requiring the failure probability to be at most δ :

$$P(|S_n - \mu| \geq \epsilon) \leq \delta$$

Chebyshev's inequality gives us the upper bound on the failure probability:

$$P(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2 n}$$

To guarantee that the failure probability does not exceed δ , we make this upper bound less than or equal to δ :

$$\frac{\sigma^2}{\epsilon^2 n} \leq \delta$$

This gives us:

$$\frac{\sigma^2}{\epsilon^2 \delta} \leq n \implies n \geq \frac{\sigma^2}{\epsilon^2 \delta}$$

Which means, to ensure S_n is an ϵ -approximation with probability at least $1 - \delta$, we need to collect at least:

$$n = \left\lceil \frac{\sigma^2}{\delta \epsilon^2} \right\rceil \text{ samples}$$

Since, n is an integer.